🥇 1️⃣ Competency 9 — Most central

> **Understand the relationship between the context of the problem and the representation of the mathematical solution**

Why this is the most important competency in this task:

- The problem presents:
  - 📷 a **three-dimensional physical construction** (stacked dice)
  - 👀 a **two-dimensional viewing perspective** (top view)
- Students must understand that:
  - *"Viewed from the top"* does **not** mean counting all faces of all dice
  - Faces that are **physically hidden** do **not** exist in the mathematical representation

👉 This task requires a **translation from a real-world spatial situation to a constrained mathematical model**.

Key ideas involved:

- Some dice exist physically but are **not visible**
- Only visible faces contribute to the mathematical count
- The context directly determines **what is included or excluded** from the calculation

This aligns precisely with Competency 9.

🥈 2️⃣ Competency 5

> **Interpret mathematical outcomes in a variety of formats and compare representations**

This problem involves multiple representations of the same situation:

- A photograph (3D object)
- A top-view projection (2D)
- An abstract mathematical model ("5 visible dice")

Students must move fluently between:

> image → spatial structure → top-view representation → numerical counting

This is not a purely computational task, but one of **representation translation**.

🥉 3️⃣ Competency 7

> **Construct and communicate explanations and arguments**

The solution process inherently involves reasoning, such as:

- Why only five dice are visible from the top
- Why the central dice is hidden
- Why only the top faces are counted

Students must **justify** their reasoning, not merely provide a numerical answer.

4️⃣ Competency 1

> **Interpret a mathematical result back into the real-world context**

Even after obtaining a numerical result, students must understand:

- What the number represents
- That it is **not** the sum of all dots on all dice
- That it is the total number of dots **visible from the top**

The mathematical result must be **re-embedded in the physical context**.

5️⃣ Competency 8

> **Recognise the extent and limits of mathematical concepts and solutions**

This task highlights that:

- Mathematical operations (such as summation) are **context-dependent**
- The solution is limited by **viewing perspective**
- Not all physically existing objects are mathematically relevant

Students must recognise that the solution is **conditionally valid**.

6️⃣ Competency 2

> **Identify whether a result makes sense given the context**

For example:

- A total exceeding the maximum possible dots on five dice would be unreasonable
- A very small total might indicate missing visible faces

This involves **reasonableness checking**, though it is not the primary focus.

7️⃣ Competency 6

> **Use real-world knowledge to make contextual judgments**

This appears in background knowledge:

- Dice are cubes
- Solid objects block the view of surfaces behind them
- Visibility is physically constrained

These real-world facts support the reasoning but are not the main assessment goal.

8️⃣ Competency 3

> **Identify the limits of the model used**

The mathematical model assumes:

- Only the top view matters
- Other perspectives are irrelevant

However, the task does not explicitly ask students to critique or extend the model, so this competency is only weakly involved.

9️⃣ Competency 4 — Least relevant

> **Use mathematical tools or computer simulations**

- No technology or simulation is required
- The task relies on **spatial reasoning and contextual interpretation**

🥇 1️⃣ 第 9 点（最核心）

> **Understand the relationship between the context of the problem and representation of the mathematical solution**

为什么最重要?

- 题目给的是：
  - 📷 **三维实物结构（骰子堆叠）**
  - 👀 **二维视角（top view）**
- 学生必须理解：
  - **“从上面看到” ≠ 所有骰子的所有点数**
  - 被遮挡的面 **在数学表示中不存在**

👉 这是一个**从现实情境 → 数学表示（可见点数）**的典型转换问题。

体现点：

- 哪些骰子“存在但不可见”
- 哪些点数在三维结构中 **不进入数学计算**
- 情境直接决定了数学对象的“可见性”

✔️ 完全命中第 9 点的核心描述。

🥈 2️⃣ 第 5 点

> **Interpret mathematical outcomes in a variety of formats; compare representations**

体现在哪里?

- 同一个结构有：
  - 实物照片（3D）
  - 顶视图（2D）
  - 抽象成“5 个可见骰子”的数学模型
- 学生需要在这些表示之间**来回切换**：
  - 图像 → 空间结构 → 顶视投影 → 数点

👉 这不是纯计算，而是**多重表示的理解与转换。**

🥉 3️⃣ 第 7 点

> **Construct and communicate explanations and arguments**

本题的解题过程本身就是一个论证过程：

例如：

- 为什么只能看到 5 个骰子?
- 为什么中间那颗骰子看不到?
- 为什么只数“顶面”的点?

学生必须**解释“为什么这样数是合理的”**，而不是只给答案。

✔️ 很适合考察“说理型”数学表达。

4️⃣ 第 1 点

> **Interpret a mathematical result back into the real world context**

即使算出了一个数字（比如 17 个点），还必须理解：

- 这个数字代表什么？
- 是“所有骰子点数之和”吗？（❌不是）
- 是“从上面能看到的点数总和”✅

👉 数学结果**必须回扣到真实观察情境。**

5️⃣ 第 8 点

> **Recognise the extent and limits of mathematical concepts and solutions**

体现为：

- 数学中的“加总”并不是无条件的
- 解的范围被**视角限制**
- 并非“看到 7 个骰子 → 数 7 个顶面”

学生要意识到：

> 数学解是 **有条件成立的**

6️⃣ 第 2 点

> **Identify whether a result makes sense given the context**

例如：

- 如果算出点数比 5 个骰子最大可能还大 → ❌不合理
- 如果算出点数太小 → 可能漏数

👉 属于**结果合理性检验**，但不是本题最核心目标。

7️⃣ 第 6 点

> **Use real-world knowledge to adjust or judge results**

体现较弱，但仍存在：

- 骰子是立方体
- 实体会遮挡
- 现实中的“看见”有物理限制

这是背景支撑，而非主要考点。

8️⃣ 第 3 点

> **Identify the limits of the model used**

模型就是：

- “顶视图只考虑可见顶面”

但题目并未要求学生反思：

- 如果从侧面看会怎样
- 模型是否忽略其他信息

👉 隐含存在，但不是显性目标。

9️⃣ 第 4 点（最不相关）

> **Use mathematical tools or computer simulations**

- 完全不需要技术工具
- 这是一个**空间想象 + 情境理解**题

✔️ 基本不涉及。